



CAPSTONE PROJECT

AI Computer Vision in Automotive Service Centers

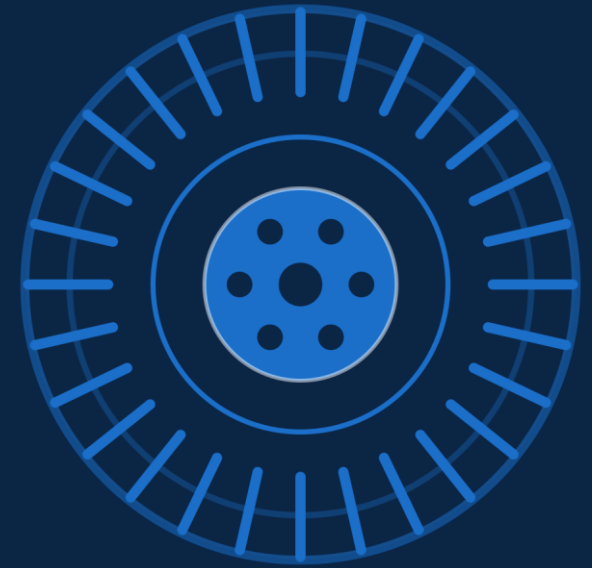
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Where Every Service Visit Begins

The Vehicle Intake Process in Automotive Service Centers



- Every vehicle entering a dealership or independent service center passes through an intake inspection before any work is quoted.
- Technicians visually assess tires, brakes, and fluids in minutes — under time pressure, and often between other bay tasks.
- Tire condition is one of the highest-frequency, highest-liability checkpoints: it drives both customer safety and after-sales revenue.
- This intake moment is the single best opportunity to convert an inspection into a sale — or to miss it entirely.

Manual Tire Inspection Is Slow, Inconsistent, and Costly



SAFETY RISK

Subjective Visual Checks

Tread-depth judgment varies technician to technician — inconsistent calls put customer safety and liability on the line.



OPERATIONS

Time Bottlenecks

Manual gauge measurement on all four tires slows intake, backing up bays during peak service hours.



REVENUE

Leakage from Missed Sales

Worn tires that go unflagged at intake become quotes never written — and sales the center never makes.

Computer Vision at the Bay Door



- A mounted or handheld camera captures each tire during intake, before the vehicle reaches the bay.
- A CNN-based image classifier scores tread condition in real time, flagging tires as New / Acceptable / Worn-Unsafe.
- Results are logged automatically to the vehicle's service record — removing guesswork from a 5-second technician glance.
- Consistent, repeatable classification standardizes the safety call across every technician and every shift.

INTAKE-TO-CLASSIFICATION FLOW



Vehicle Arrives

Intake bay camera positioned at tire height



CV Capture & Scan

Tread pattern imaged in real time



Classify Condition

New / Acceptable / Worn-Unsafe



Log to Service Record

Result attached to the vehicle file

System Cost: Itemized Breakdown

Estimated cost per service center, scaled to a premium hardware and integration tier.



Hardware & Connectivity

480,000 EGP

- Camera system (multi-angle, industrial)

200,000 EGP

- Edge AI compute unit

200,000 EGP

- Networking & cabling

80,000 EGP



Installation & Site Prep

120,000 EGP

- Mounting, structural & electrical work

120,000 EGP



Software & Integration

300,000 EGP

- CV software license/deployment

180,000 EGP

- ERP integration

120,000 EGP



Enablement & Contingency

100,000 EGP

- Staff training

60,000 EGP

- Contingency (10%)

40,000 EGP

TOTAL PER SITE: 1,000,000 EGP

× 15 SITES = 15,000,000 EGP

Bootstrap Data: Kaggle Tyre Quality Dataset

Before any service-center images existed, the classifier was bootstrapped on this public dataset.

“Tyre Quality Classification”

kaggle.com/datasets/warcoder/tyre-quality-classification

- 1,854 digital tyre images across 2 classes: Defective and Good condition.
- Originally published as “Digital images of defective and good condition tyres,” Mendeley Data, 2023.
- Licensed CC BY 4.0 (Attribution 4.0 International) — free to use and adapt with credit.
- Used here to bootstrap the classifier before fine-tuning on real service-center intake images (Section III).



Good — deep tread



Defective — worn tread

Dataset Selection & Data Types

Public Kaggle Tire Quality/Wear Classification dataset — chosen to keep the project fully compliant with academic and institutional data-use policies, with no proprietary or customer data required at this stage.



INPUT

Unstructured Data

Raw tire tread images — the unprocessed visual input captured at vehicle intake and fed into the classification model.



MODEL OUTPUT

Structured Categorical Data

The AI model converts each image into a discrete class label: Safe or Worn — ready for logging and reporting.



MODEL OUTPUT

Numerical Confidence Scores

Each classification carries a confidence score (0–100%), giving service teams a measure of prediction certainty.

Data Processing Workflow

- Model classification outputs (label + confidence score) are captured for every simulated intake event.
- Python and Pandas, run inside a Jupyter Notebook on the Anaconda environment, structure and clean these outputs.
- The pipeline generates a structured CSV log simulating daily service center intake operations — one row per vehicle.
- This CSV becomes the single source of truth feeding both the KPI calculations and the Power BI dashboard.

PROCESSING PIPELINE



CV Model Output

Label + confidence score per tire



Python & Pandas

Jupyter Notebook, Anaconda environment



Structured CSV Log

One row per simulated intake event



Ready for Analytics

Feeds KPIs and the Power BI dashboard

Key Performance Indicators

Three business metrics translate model output into operational and financial impact.



OPERATIONAL EFFICIENCY

Intake Time Reduction

Average minutes saved per vehicle at intake compared to fully manual tread-depth measurement.



SAFETY COMPLIANCE

Defect Detection Rate

Share of truly worn/unsafe tires correctly flagged by the model, validated against technician review.



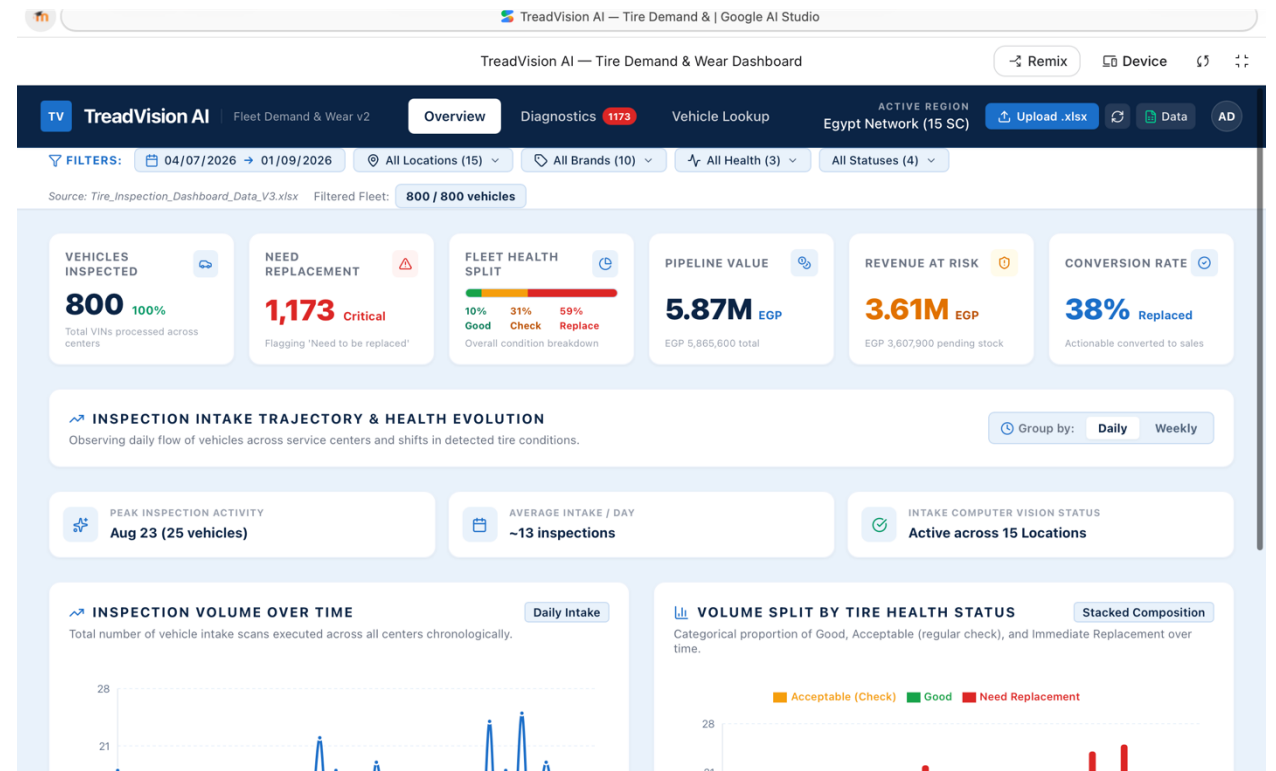
FINANCIAL IMPACT

Missed Revenue Capture / ROI

Value of tire-replacement sales recovered by catching worn tires that manual checks would have missed.

From Classification to a Live Ops Dashboard

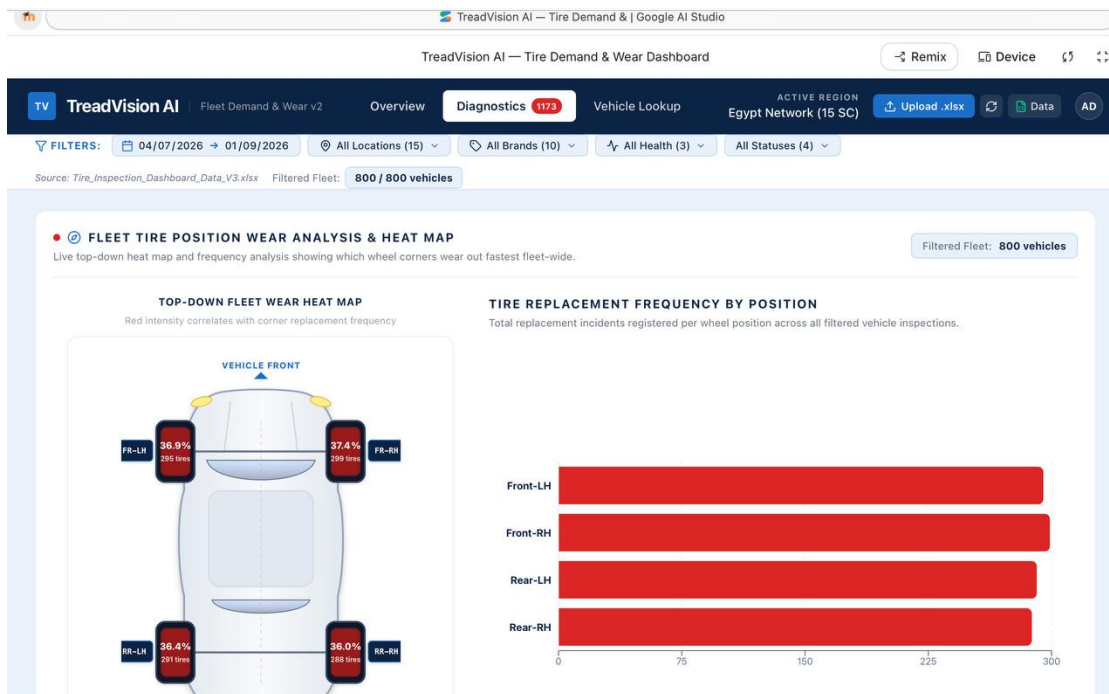
- Inspection Accuracy — model precision vs. technician-verified spot checks.
- Vehicle Throughput — intake volume and average cycle time per bay.
- Worn-Tire Flag Rate — share of intakes classified Worn-Unsafe, by location.
- Quote Conversion — flagged tires converted into completed sales.



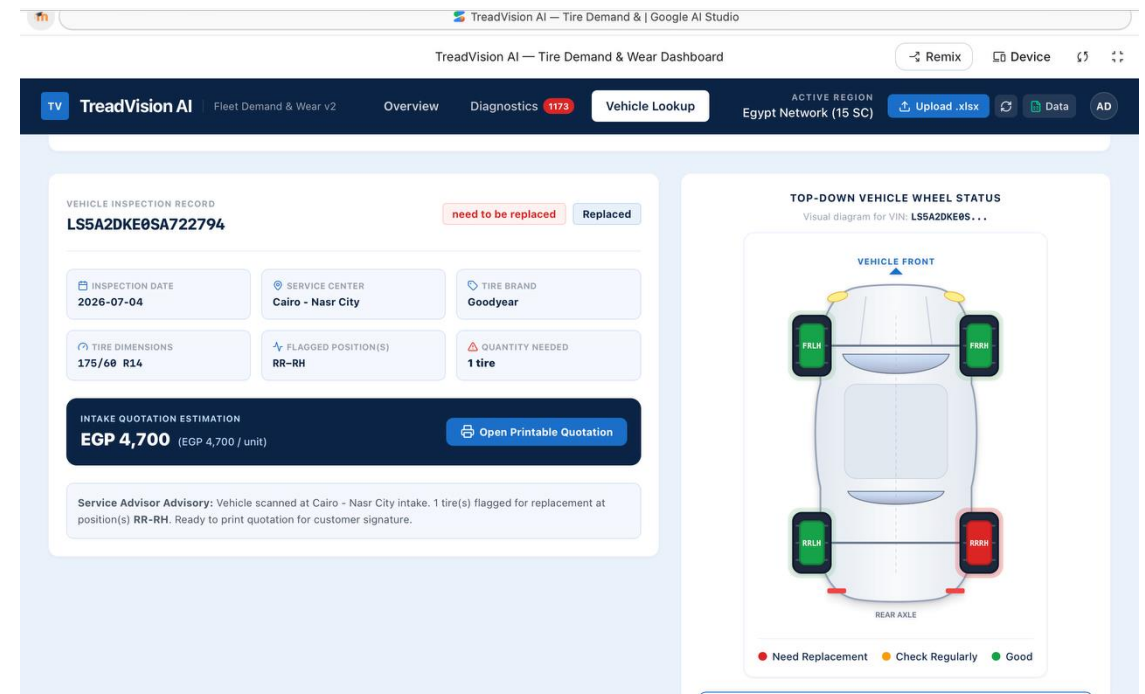
Live dashboard, built on the classification pipeline's structured output.

From Fleet Pattern to a Single Quotation

Diagnostics finds where the fleet-wide problem is; Vehicle Lookup turns one flagged tire into a printable customer quotation.



Diagnostics: fleet-wide heat map



Vehicle Lookup: one VIN, one quotation

Closing the Loop: Detection to Dollars



The ERP link is what turns a correct classification into a booked service line — automatically checking stock and drafting the quote before the advisor even walks over.

↑ Tire Sales

Fewer missed replacement opportunities at intake

↓ Inspection Time

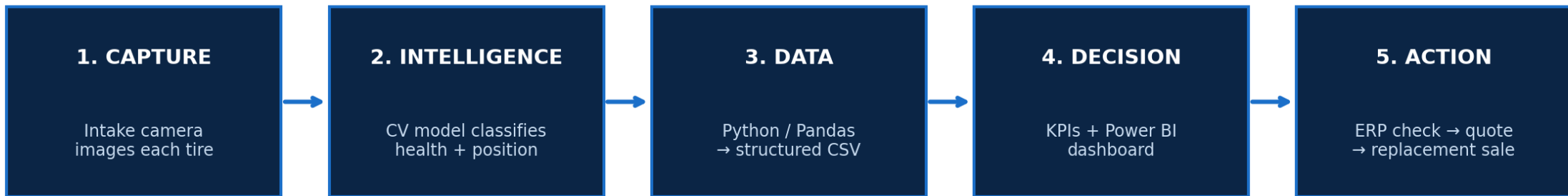
Faster, standardized intake per vehicle

↑ Safety Consistency

Every tire assessed against the same standard

The TreadVision AI Operating Framework

Every component built so far maps onto one continuous, five-layer pipeline — from the camera at intake to the completed sale.



Each layer has a single owner and a single output — Capture produces an image, Intelligence produces a label, Data produces a row, Decision produces a KPI, Action produces a sale — which is what makes the framework auditable end-to-end.

Simulation Design: Manual vs. CV-Automated

The 800-vehicle V3 dataset is treated as the CV-automated ground truth. A manual-inspection baseline is simulated on the same population using two explicit, labeled assumptions:



ASSUMPTION 1

Manual Detection Rate: 62%

A rushed, subjective visual check catches roughly 62% of truly worn/unsafe tires — versus the CV model's near-complete recall on this labeled dataset.



ASSUMPTION 2

Manual Time: 20 min/vehicle

Manual four-tire tread-depth measurement plus a visual check takes about 20 minutes, versus 2 minutes for camera capture and auto-classification.



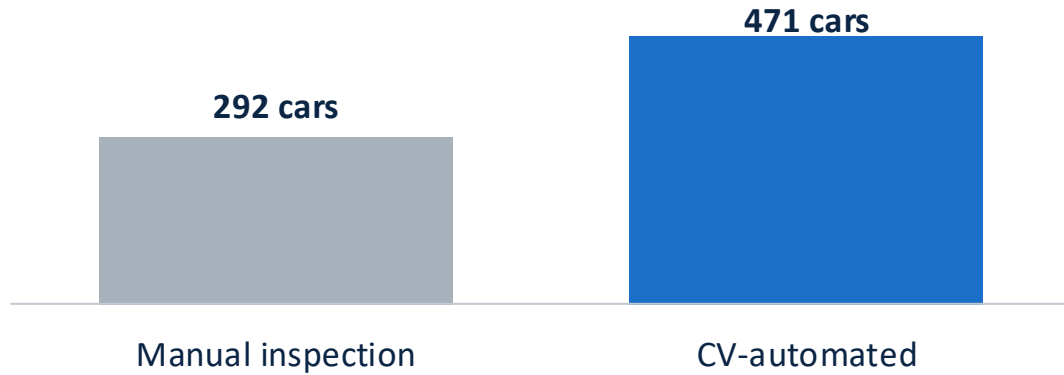
HELD CONSTANT

Downstream Conversion Process

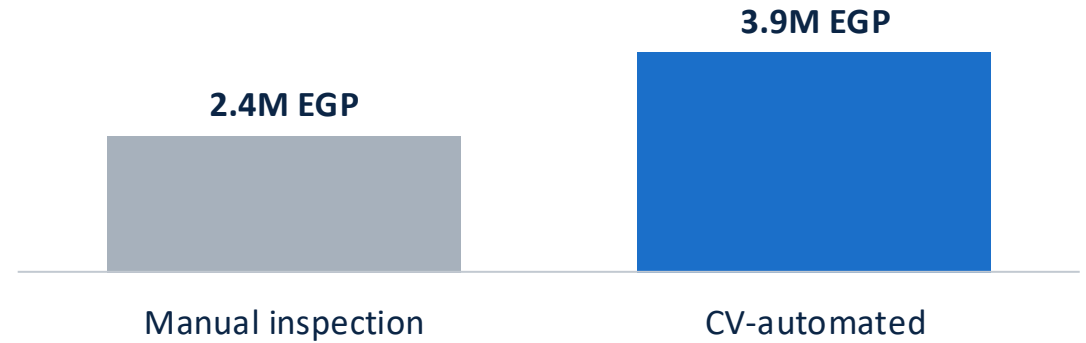
Once a tire is flagged, the stock-check → quote → replace funnel behaves identically in both scenarios — isolating detection accuracy as the only variable.

Simulation Results

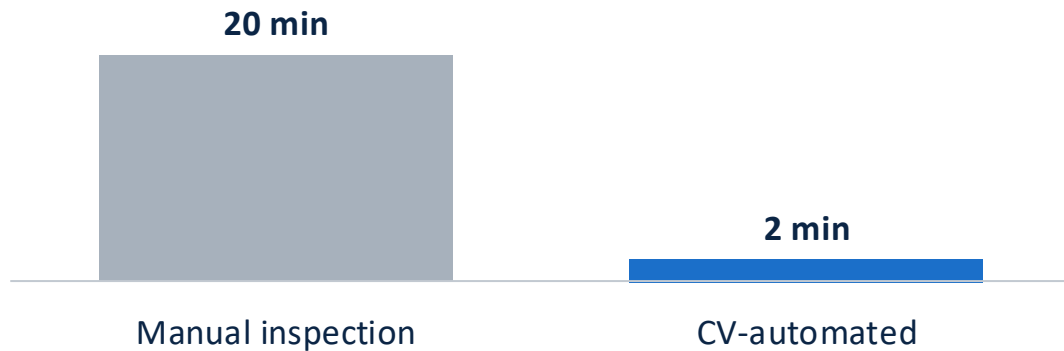
Worn vehicles correctly flagged, of 800 inspected



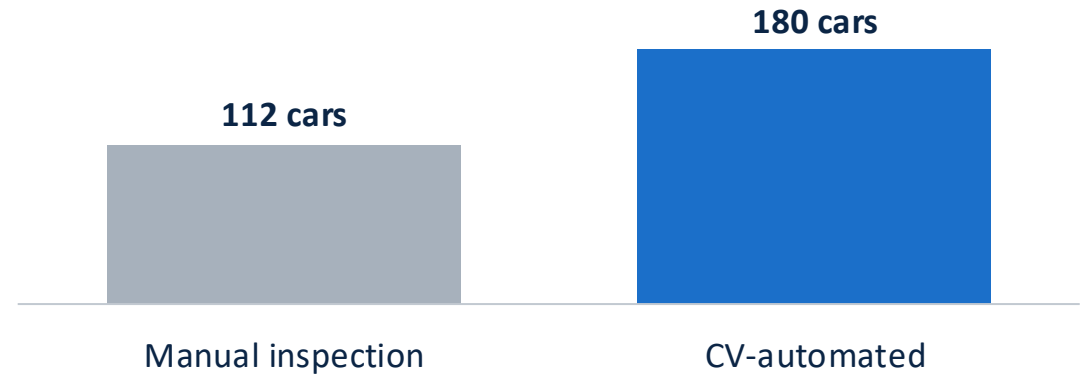
Potential revenue identified, 60-day window



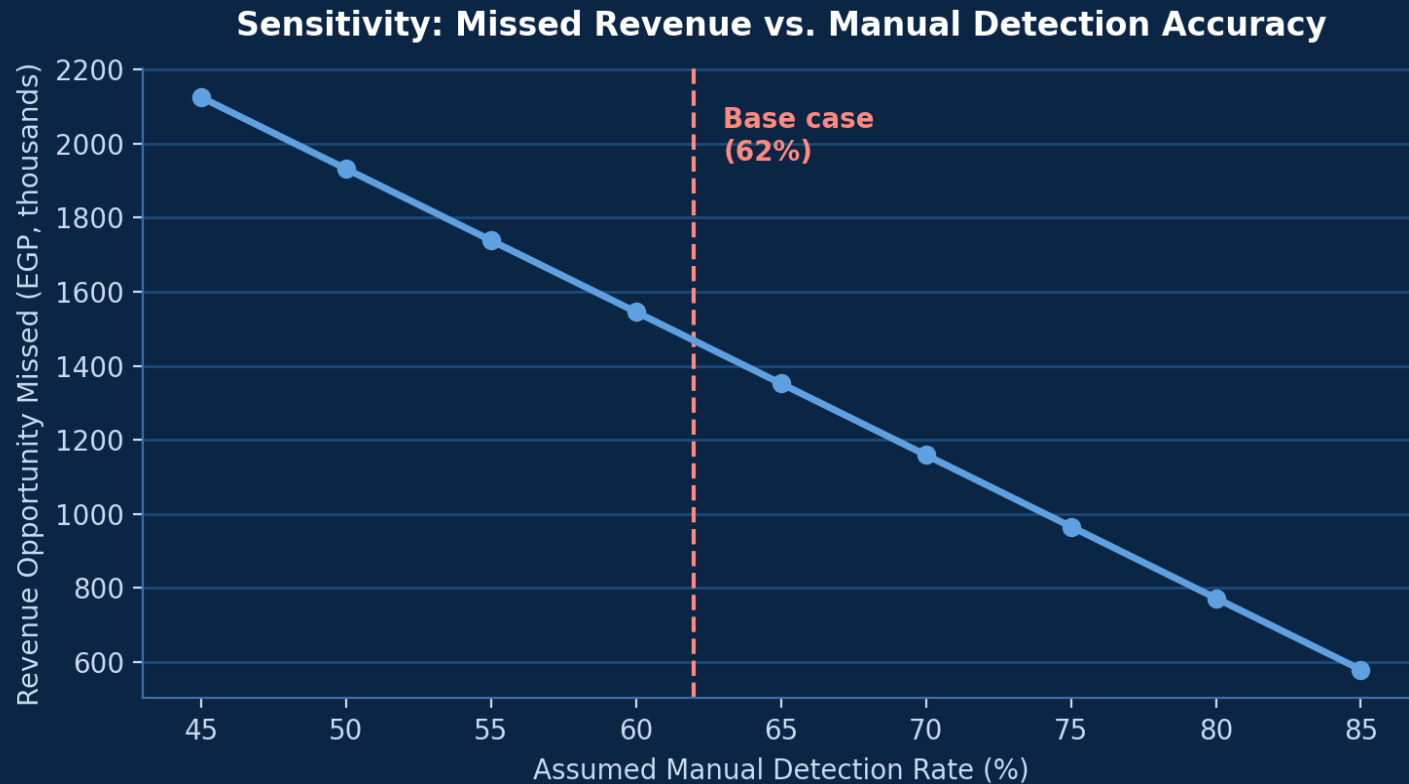
Average inspection time per vehicle



Completed tire replacements, 60-day window



Sensitivity & Headline Takeaway



ASSUMED MISSING SALES

179 cars

1.47M EGP

SURE MISSING SALES

68 cars

0.59M EGP

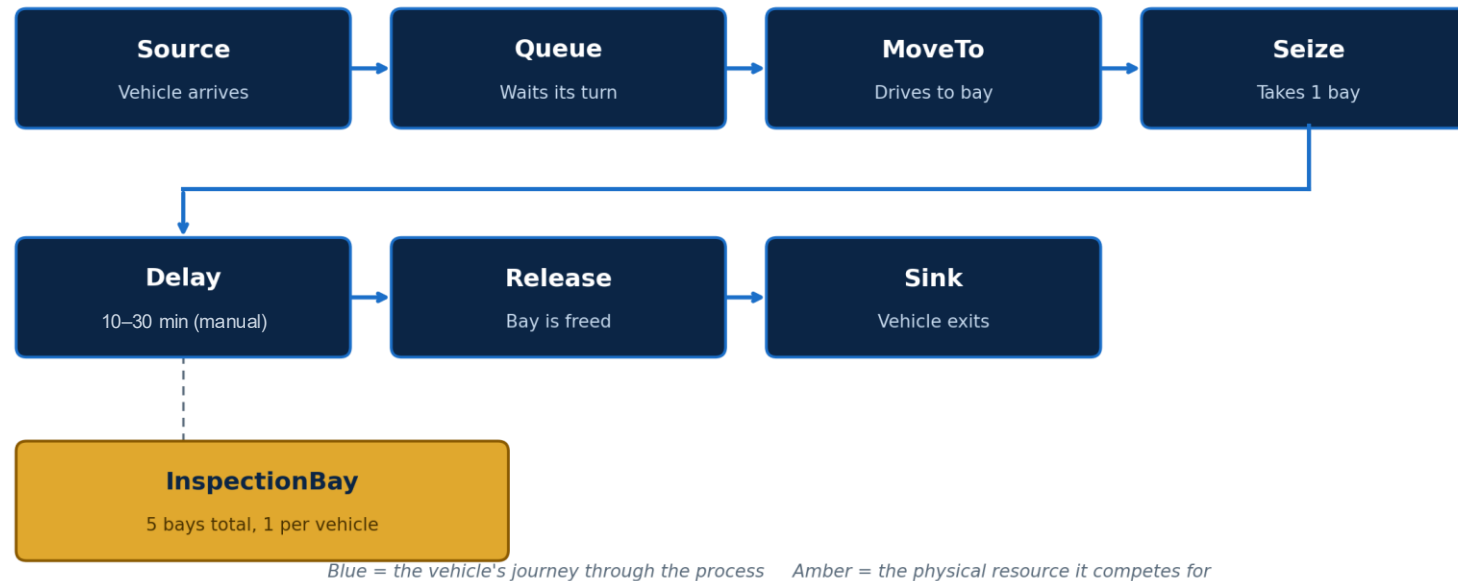
TIME SAVED, FLEET-WIDE

~1,460 hrs/yr

inspection time saved by moving to CV

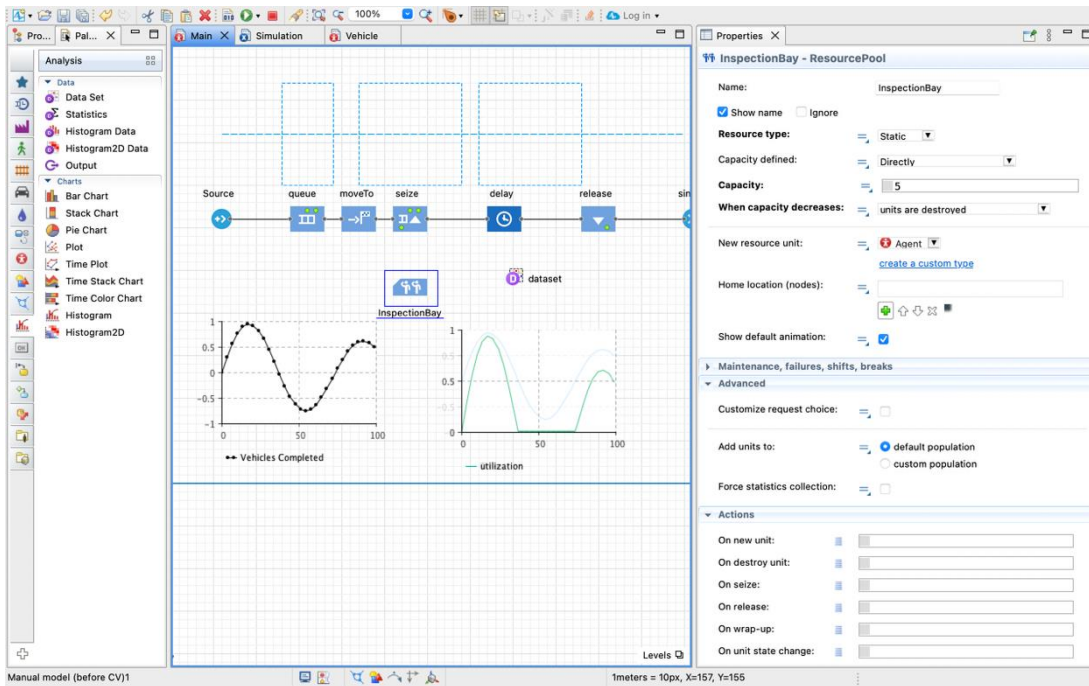
AnyLogic Implementation: Process Flow

The queue simulation in Section VII was also built directly in AnyLogic, using the Process Modeling Library, to validate the Python model interactively.

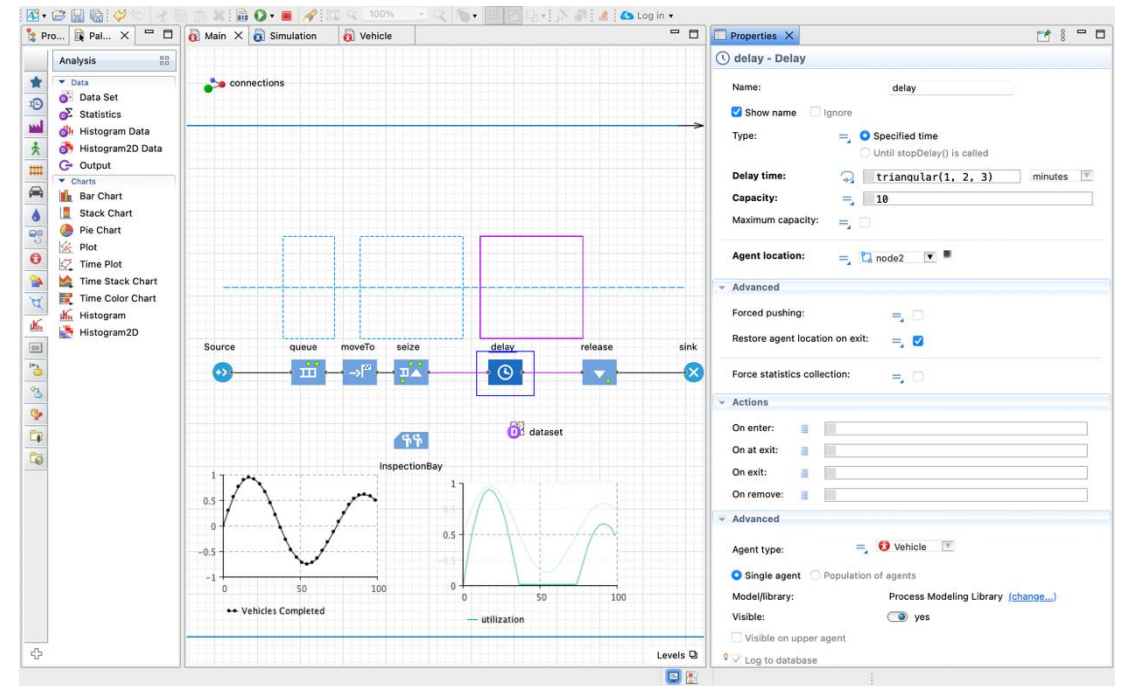


Each vehicle flows left to right through six process blocks, competing with other vehicles for the InspectionBay resource pool (5 bays).

The AnyLogic Build: Model Settings

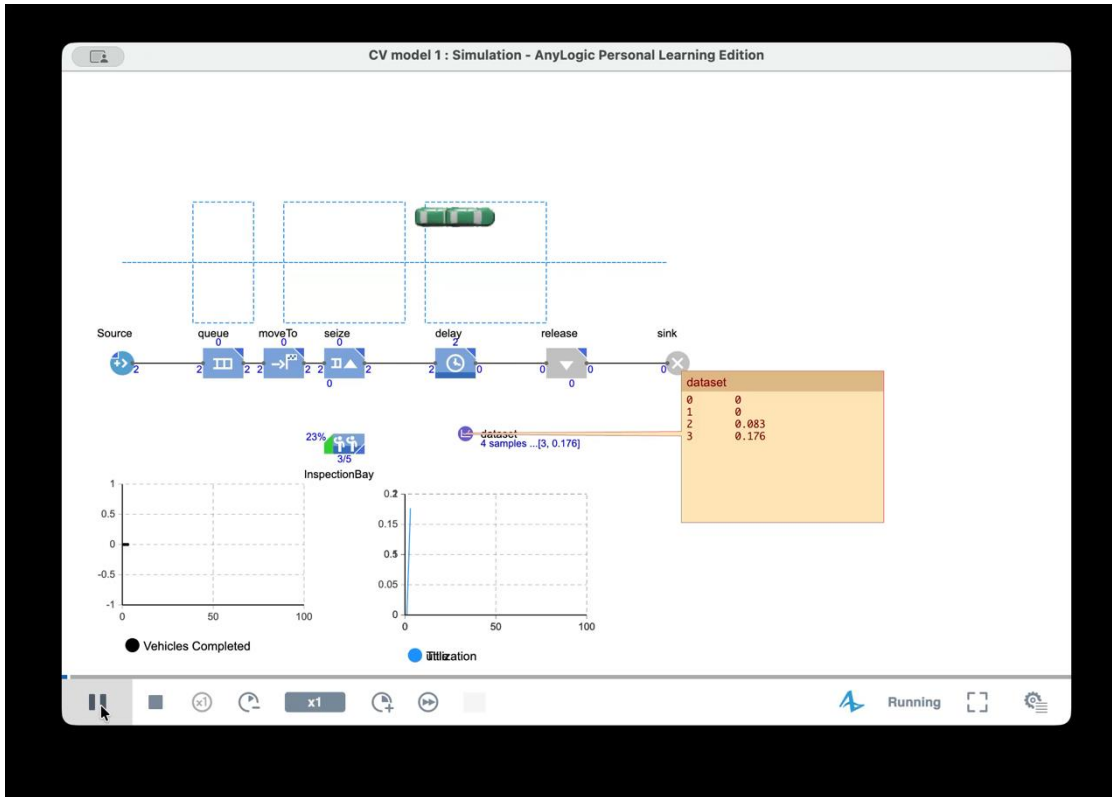


Inspection Bay setting

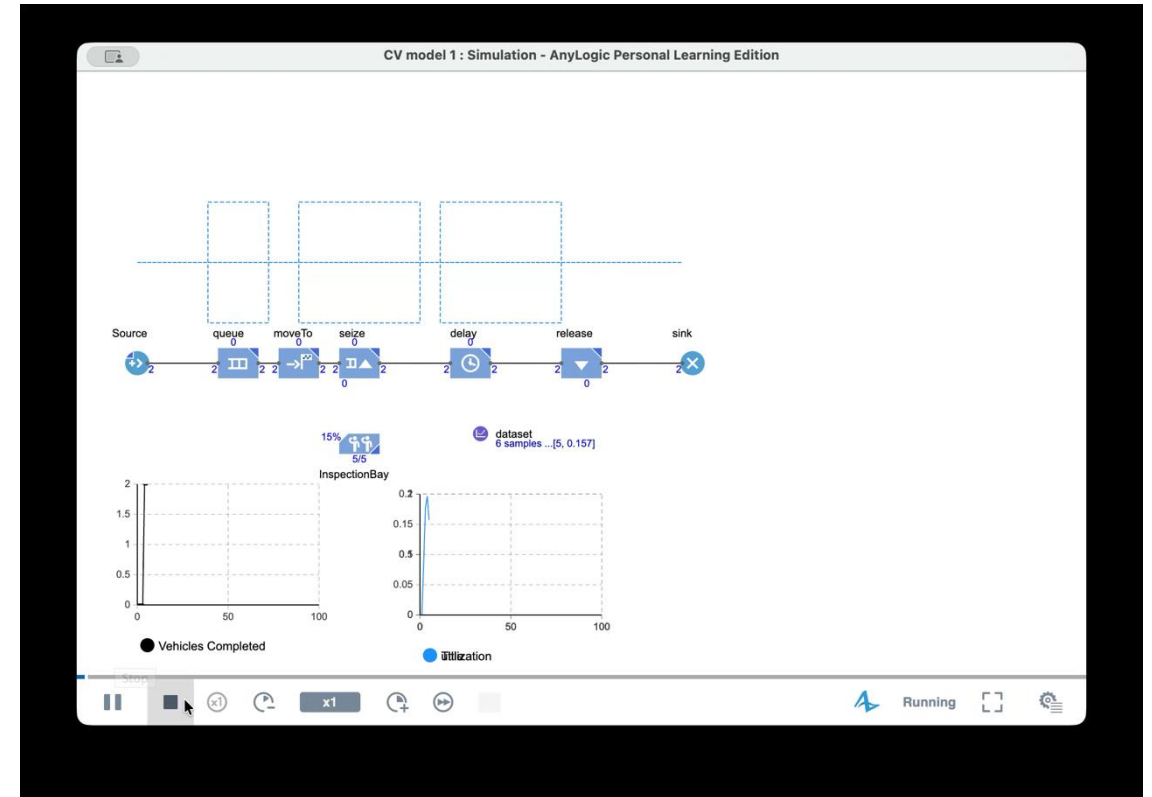


CV time setting

Manual vs CV Inspection, Side by Side



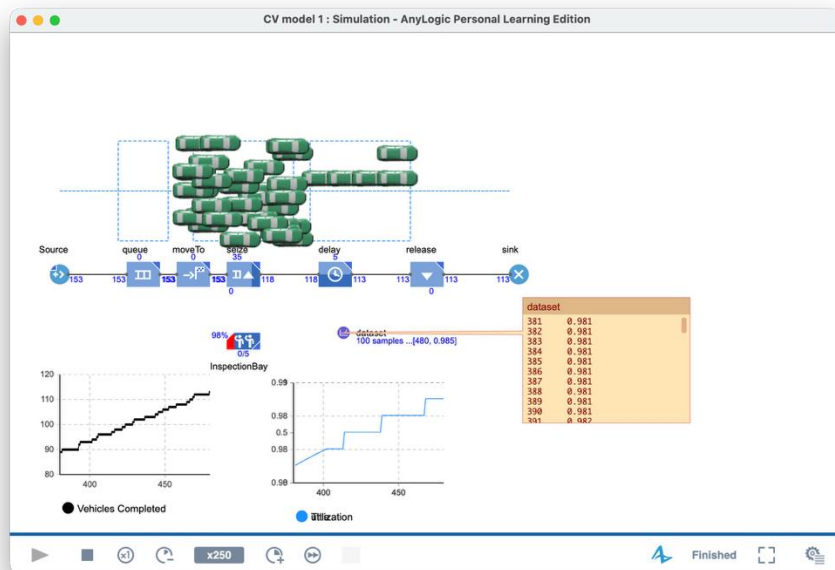
BEFORE — Manual Inspection



AFTER — CV-Automated

Proof: Manual vs. CV-Automated

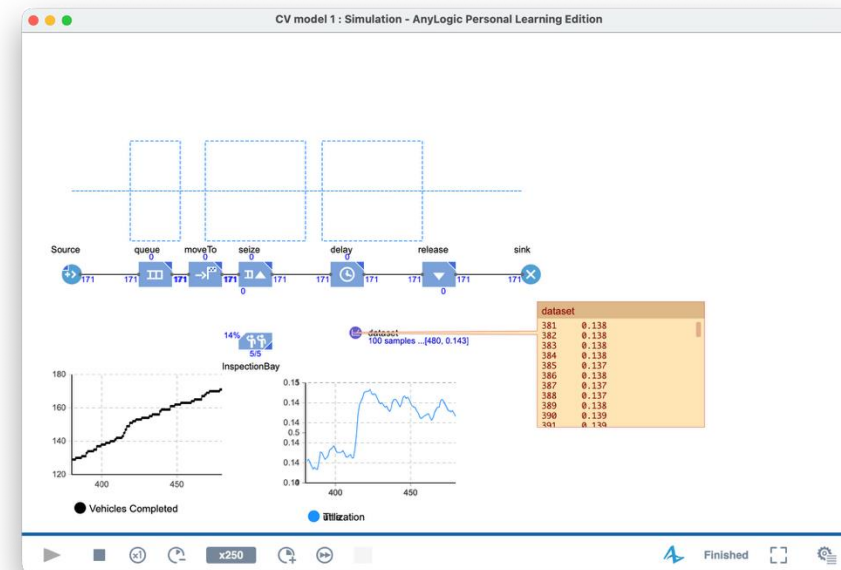
Two independent AnyLogic runs, each simulating a full business day — identical setup, only the inspection method differs.



BEFORE — Manual Inspection

113 → 171

Vehicles fully inspected (+51%)



AFTER — CV-Automated

40 → 0

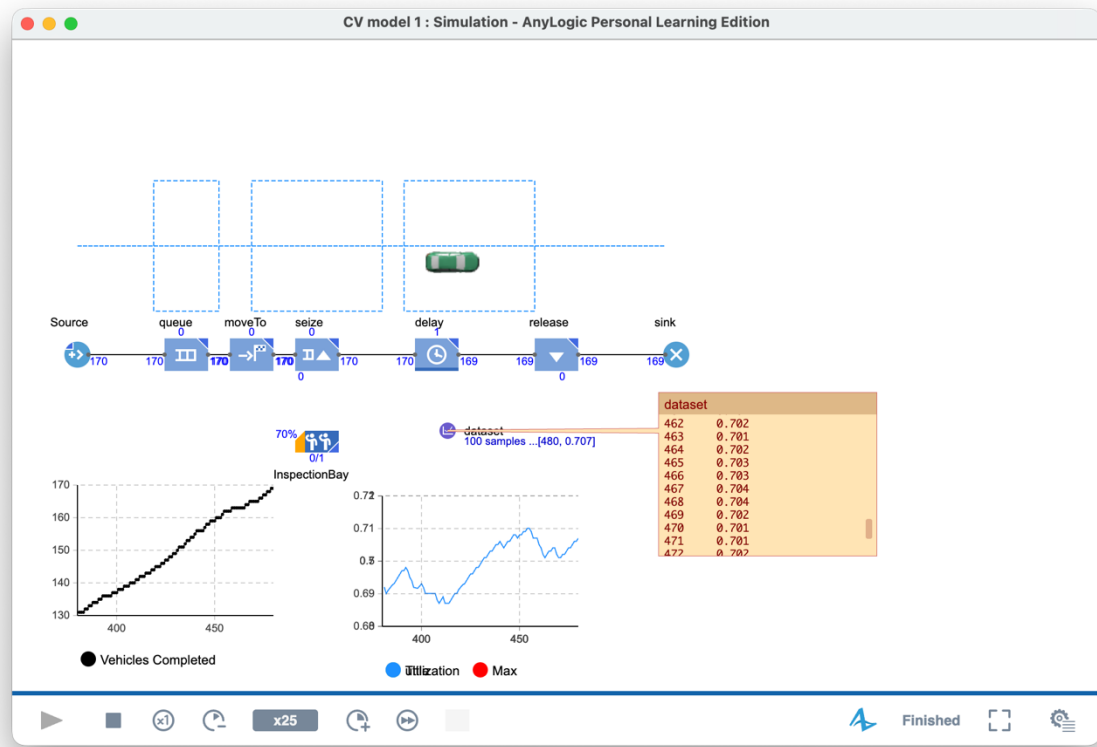
Vehicles still stuck at closing time

98% → 14%

Inspection bay utilization

Scenario Test: One Inspector Instead of Five

Same CV-automated AnyLogic run reducing the labors from 5 to only 1.



5 → 1

Inspection technicians per shift (~80% labor)

171 → 169

Vehicles inspected per day — only 2 fewer

14% → 71%

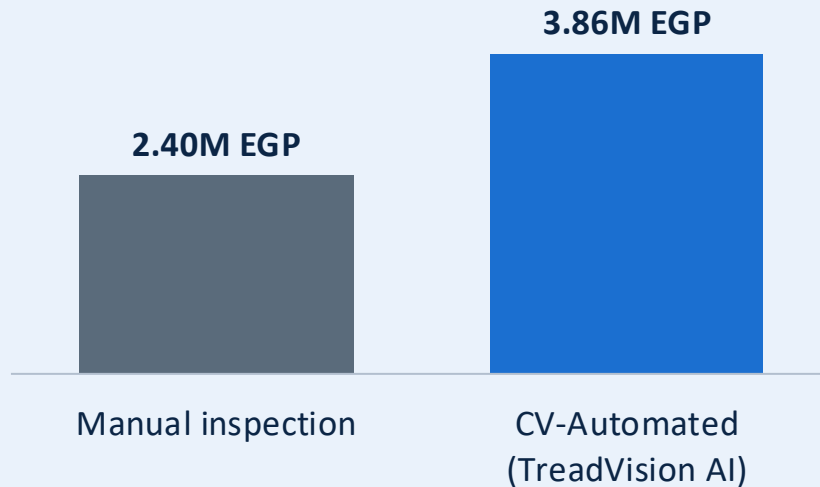
Inspector utilization — busy but healthy, no queue

Revenue Impact & Broader Benefits

TIER 1 — MEASURED FROM THE SIMULATION

≈ 7.8M EGP / year

Fleet-wide revenue recovered by catching worn tires that manual inspection misses entirely.



Potential revenue identified over the 60-day dataset window, Manual vs. CV-Automated.

ADDITIONAL BENEFITS — REAL, NOT QUANTIFIED HERE

Labor Redeployment

1 inspector now does the work of 5 (previous slide) — 80% of inspection labor freed for upselling and other services, or taken as cost savings.

Faster Tire Turnover

Tires move from warehouse to installation faster, reducing inventory holding cost.

Improved Cash Flow

Faster sales conversion accelerates the cash conversion cycle.

Customer Satisfaction

Shorter waits and fewer turned-away customers support stronger word-of-mouth referrals.

These effects are directionally real but require data this project doesn't yet have (labor cost, inventory turnover, customer surveys) to quantify — included here as an honest scope boundary, not a hidden assumption.

Revenue & ROI: The Business Case

The investment pays for itself in under two years, then compounds into a strong 5-year return.

Total Investment (CapEx)	15,000,000 EGP
Annual Depreciation (5-yr)	– 3,000,000 EGP/yr
Annual OpEx	– 1,095,000 EGP/yr
Gross Annual Revenue Benefit	+ 8,932,974 EGP/yr
Net Annual Cash Benefit	7,837,974 EGP/yr

PAYBACK PERIOD

≈ 1.9 years

(~23 months)

ANNUAL ROI (CASH)

≈ 52.3%

return per year

5-YEAR CUMULATIVE ROI

≈ 161%

over the depreciation life

5-YEAR NET PROFIT

≈ 24.2M EGP

after recovering the investment

The project pays back its full investment in under 2 years, then returns 161% over its 5-year depreciation life — a financially sound, defensible investment.



Thank You

Questions and Discussion

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